Rakshit Rangaprasad

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EDUCATION

Tufts University **Medford, MA**
Bachelor of Science in Computer Engineering *Expected May 2028*

- **Relevant Coursework**: Linear Algebra, Multivariable Calculus, Embedded Systems, Machine Learning, Digital Logic, Data Structures, Discrete Mathematics, Computer Organization (In Progress), Intro to Circuits (In Progress)

SKILLS, ACTIVITIES & INTERESTS

Languages: C, C++, Python, Java, TypeScript, JavaScript, MATLAB, (HDL: SystemVerilog)
Embedded & Hardware: ESP32, STM32, KiCad (PCB design), I2C, SPI, UART, PWM, ADC, Arduino Framework
Tools & Software: Git, PlatformIO, Linux, WebSockets, CAD (Onshape), Surfer (Waveform Debug Tool), Oscilloscope
Interests: Tennis, South Indian Cuisine, Cooking, Mark Rober, Hackathons (Hackachusetts, HackMIT, HackHarvard), Chess

TECHNICAL PROJECTS

Autonomous Robotic Vehicle Platform, "Blues Clues" (for EE31: Junior Design Project)
C++, ESP32-S3, PlatformIO, Arduino Framework, WebSockets, Onshape, KiCad *2026*

- Developed firmware on an ESP32-S3 using **C++** in PlatformIO to control dual DC motors via an L293D H-bridge, optimizing **PWM** frequency to achieve a 10% duty cycle threshold.
- Engineered an **IR obstacle-sensing circuit** conditioned for ESP32 **ADC** logic and calibrated count/rev magnetic encoders to ensure no speed mismatch between left and right wheels.
- Developed a chassis in **Onshape (CAD)** with mounting considerations for sensors, and a **PCB** in **KiCad**.
- Debugged PWM, encoder, and sensor signals using **oscilloscopes** to validate circuit behavior.

"Porta-Pot" (for EE14: Embedded Systems)
C++, ST Nucleo L432KC, PlatformIO, 2026

- Implemented **nRF24L01 transceivers** for wireless SPI communication, using **C++**, between two microcontrollers.
- Integrated **ADC** joystick inputs and **PWM** motor control using a DRV8835 motor driver.
- Designed and assembled a **laser-cut wooden chassis** for a remote-controlled plant carrier.

Single Cycle CPU in SystemVerilog (for EE25: Computer Organization)
SystemVerilog, Surfer (Waveform Debugging Tool) *2026*

- Designed a 64-bit single-cycle CPU for LEGv8 in **SystemVerilog** with an ALU, register file, control unit, and memory.
- Implemented arithmetic, logical, load/store, and branch instructions.
- Created a simulation testbench for **Surfer** and debug outputs to inspect instructions, registers, and memory.

PROFESSIONAL EXPERIENCE

Onshape by PTC **Boston, MA**
Software Engineer Intern *Jun 2026 - Aug 2026*

- Reduced locale update times by **80%** by engineering a locale-selection display in the initial onboarding flow.
- Diagnosed critical bugs across **TypeScript, Angular, and Java**, improving application stability and user experience. Authored robust JUnit tests to validate complex **CAD** UI changes (i.e., query updates, feature selection).
- Reviewed engineers' **Git** pull requests for accuracy and manually tested for errors.

Project Oath **Remote**
Founding Technical Lead *Apr 2026 - Present*

- Developed Project Oath, an online credit-based mentorship exchange connecting pre-med college students, high-school mentees, and medical student mentors.
- Contributed to building and scaling a nationwide network spanning 91 high schools, 57 colleges/universities, and 22 medical schools, including Harvard, Yale, Cornell, and Brown.

EXTRACURRICULAR EXPERIENCE

Ascend Global Leaders at Tufts University
Co-Founder *Sep 2025 - Present*

- Founding member of Tufts Ascend, a chapter of Ascend New England. Organizing panel and networking events.

Hindu Students Association
Treasurer *Sep 2024 - Present*

- Manage budget and spending for HSA, funding events like Diwali, Holi, and internal executive board events.